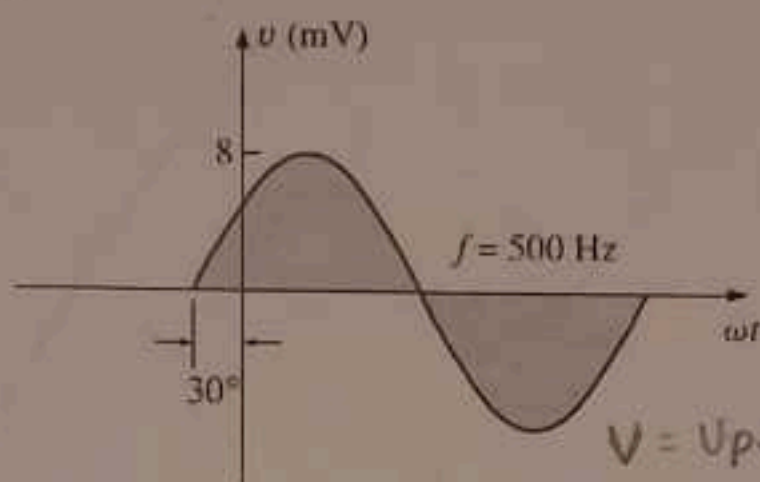


1. Encontre a expressão para as senóides mostradas abaixo:

a)

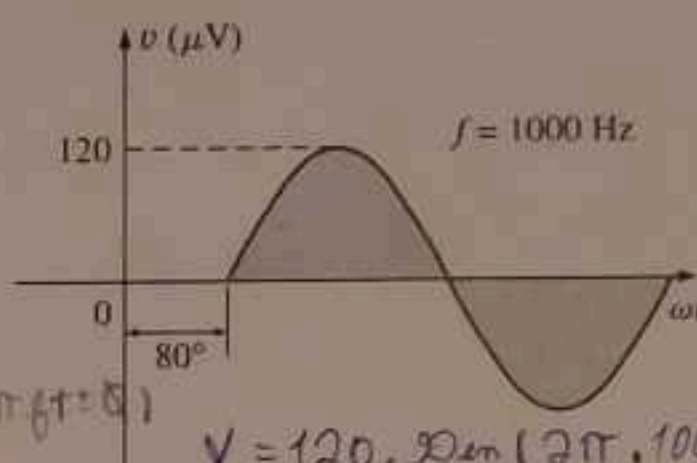


$$V = V_p \cdot \sin(2\pi f t + \phi)$$

$$v = 8 \cdot \sin(2\pi(500 \text{ Hz})t + 30^\circ) \text{ mV}$$

$$v = 8 \cdot \sin(1000\pi t + 30^\circ) \text{ mV}$$

b)



$$V = 120 \cdot \sin(2\pi \cdot 1000 \text{ Hz} \cdot t - 80^\circ)$$

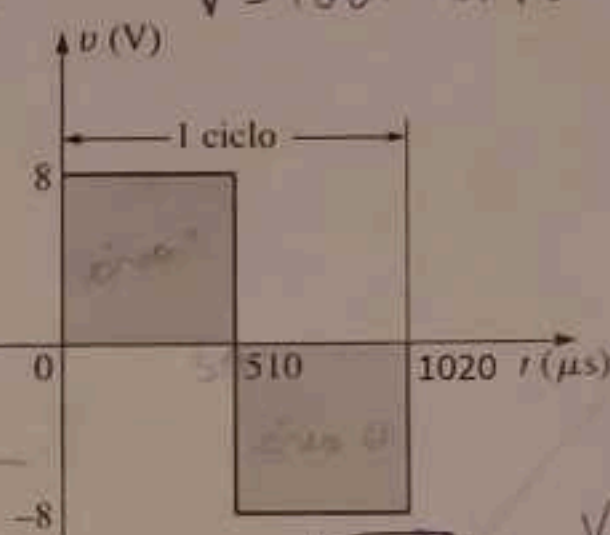
$$v = 120 \cdot \sin(2000\pi t - 80^\circ) \mu\text{V}$$

2. a) Encontre o Valor Médio do sinal ao lado.

- b) Encontre o Valor Eficaz do sinal ao lado.

$V_m = \text{valor médio}$

$$a) V_m = \frac{(8 \text{ V})(510 \mu\text{s}) + (-8 \text{ V})(510 \mu\text{s})}{1020 \mu\text{s}} \Rightarrow V_m = 0$$



$$b) V_{RMS} = \sqrt{\frac{(8 \text{ V})^2(510 \mu\text{s}) + (-8 \text{ V})^2(510 \mu\text{s})}{1020 \mu\text{s}}}$$

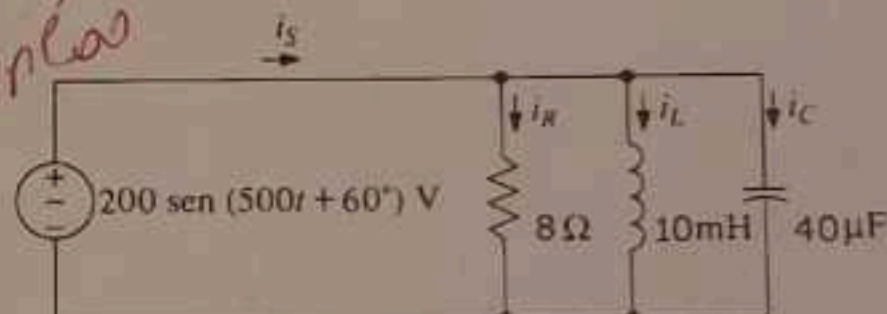
$$V_{RMS} = \sqrt{64 \text{ V}^2} \Rightarrow V_{RMS} = 8 \text{ V}$$

3. a) Encontre I_s para o circuito ao lado.

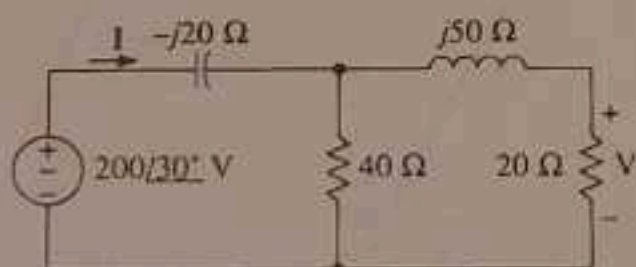
$$I_s = 30,99 \angle 4,78^\circ \text{ A}$$

- b) Encontre I_L .

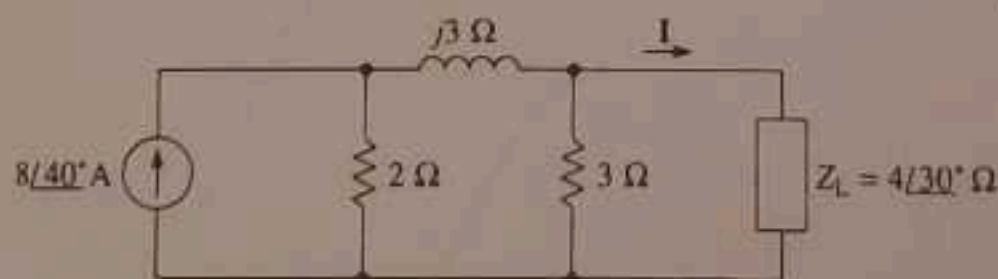
$$I_L = 28,28 \angle +20^\circ \text{ A}$$



4. Qual a corrente I e a tensão V do circuito abaixo:



5. Qual a corrente I do circuito abaixo:



$$I = 1,41 \angle -19,49^\circ \text{ A}$$

Disciplina: Eletricidade CA Prof.: Morais

Data: 20/09/23

Nota: _____

FOLHA RESPOSTA DE AVALIAÇÃO

3.a)

$$Z_T = \frac{1}{\frac{1}{8 \angle 0^\circ} + \frac{1}{5 \angle 90^\circ} + \frac{1}{50 \angle -90^\circ}}$$

$$Z_T = 4,56 \angle 55,22^\circ \Omega$$

$$E = 100\sqrt{2} \angle 60^\circ$$

$$I_s = \frac{E}{Z_T}$$

$$I_s = \frac{100\sqrt{2} \angle 60^\circ}{4,56 \angle 55,22^\circ} \Rightarrow$$

$$I_s = 30,99 \angle 4,78^\circ \text{ A} //$$

b) $I_L = \frac{E}{Z_L}$

$$I_L = \frac{100\sqrt{2} \angle 60^\circ}{5 \angle 90^\circ} \Rightarrow$$

$$I_L = 28,28 \angle -20^\circ //$$

• $Z_R = 8 \Omega //$

$Z_L = \omega L \angle 90^\circ$

$Z_L = (500 \text{ rad/s})(40 \text{ mH}) \angle 90^\circ$

• $Z_L = 5 \angle 90^\circ \Omega //$

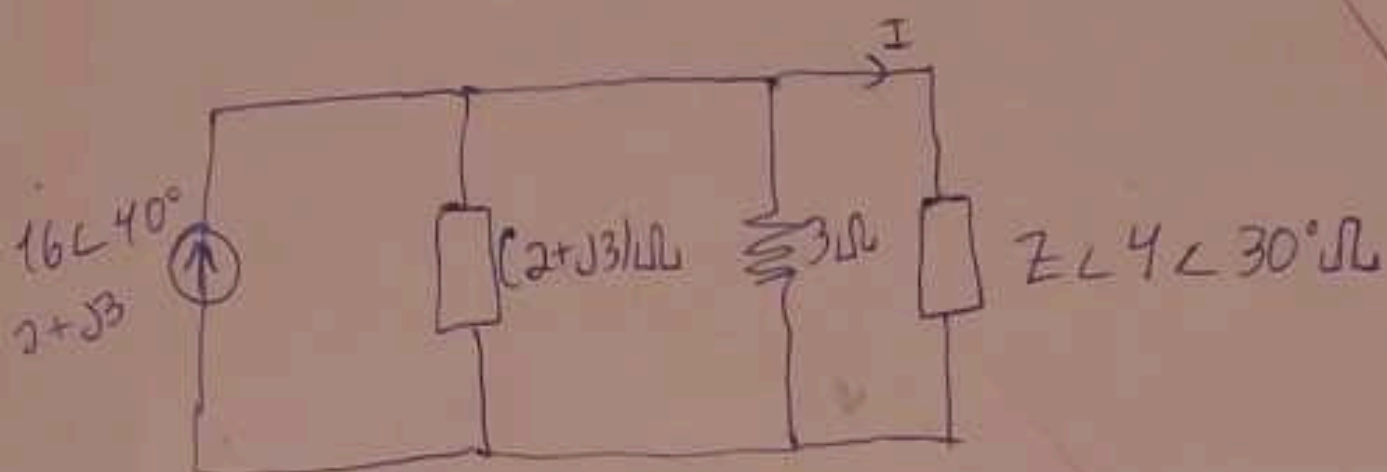
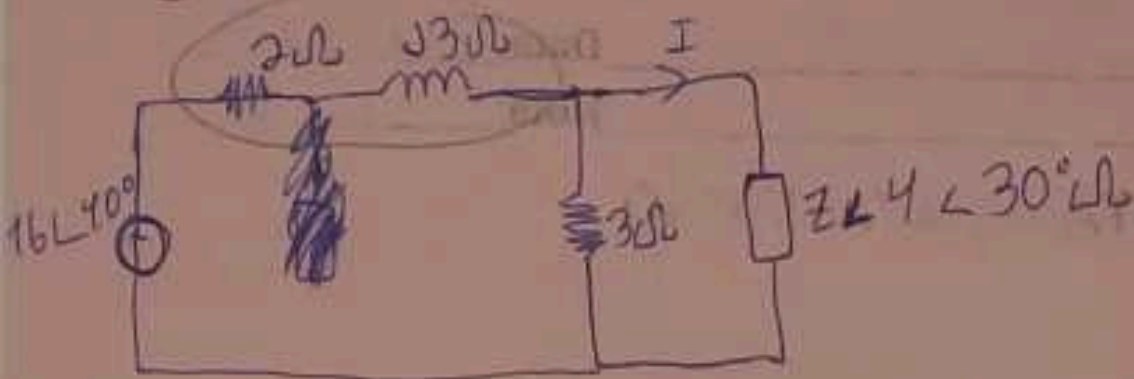
$Z_C = \frac{1}{\omega C} \angle -90^\circ \Omega$

$Z_C = \frac{1}{(500 \text{ rad/s})(40 \mu\text{F})} \angle -90^\circ$

• $Z_C = 50 \angle -90^\circ \Omega //$

A Peneira

5.



$$Z_T = 1,27 \angle 26,82^\circ \Omega$$

$$I_T = \frac{16 \angle 40^\circ}{2 + j3} \text{ A}$$

$$I_T = 4,44 \angle -16,31^\circ \text{ A}$$

$$I = I_T = \frac{Z_T}{Z_L}$$

$$I = (4,44 \angle -16,31^\circ \text{ A}) \frac{(1,27 \angle 26,82^\circ \Omega)}{(4 \angle 30^\circ \Omega)}$$

$$I = 1,41 \angle -19,49^\circ \text{ A} //$$

